

Smart Energy and Climate Management System Abstract

In this project, a simple **smart home system** has been designed and simulated, enabling control of three main sections of the house: **lighting (lamp)**, **heating**, and **cooling**. The system is implemented using an **ESP32 board** and the **Blynk platform**. Through the Blynk application, the user can control the status of each output. The entire circuit is simulated in the **Wokwi environment**, making it executable without the need for real hardware.

Introduction

The **Internet of Things (IoT)** is one of the most important fields of technology, where physical devices and objects are connected to the internet, allowing remote monitoring and control. A key application of IoT is **smart homes**, where systems such as lighting, heating, cooling, and security are intelligently managed.

The goal of this project is to design a simple smart home system capable of controlling three main sections:

- **Lighting**
- **Heating**
- **Cooling**

Blynk Platform

Blynk is a powerful platform for building IoT project dashboards. With Blynk, you can:

- Control devices remotely
- Monitor sensor status
- Build user interfaces with buttons, displays, and more

In this project, Blynk is used to control three outputs:

- **V0 → Lighting (Lamp)**
- **V1 → Heating**
- **V2 → Cooling**

ESP32 Board

The **ESP32** is a powerful microcontroller with built-in **WiFi** and **Bluetooth** capabilities. It features multiple *GPIO* pins, making it highly suitable for IoT projects. In this project, the ESP32 is responsible for controlling LEDs and communicating with Blynk.

Circuit Design

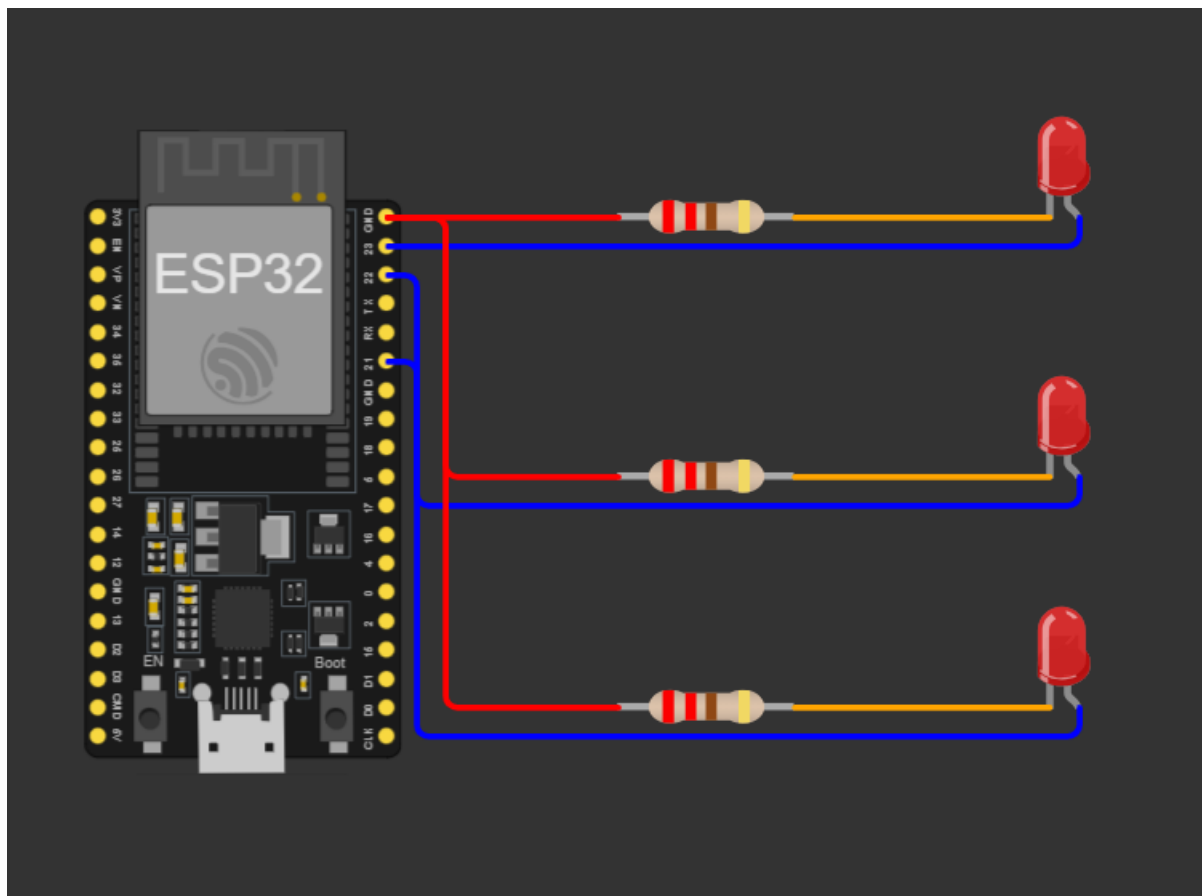
Three LEDs are used as outputs in this project:

- **GPIO23 → Lighting (Lamp)**
- **GPIO22 → Heating**
- **GPIO21 → Cooling**

LED connections:

- Positive pin → LED → Resistor → Corresponding *GPIO* pin
- Negative pin → GND

(Circuit diagram provided in Wokwi simulation)



Conclusion

This project demonstrates a simple smart home system that uses **ESP32** and the **Blynk platform** to control three main household outputs. It serves as a practical example for learning IoT concepts, remote control, and hardware simulation in the **Wokwi environment**.

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